PRACTICAL 2

C++ Program: Student Class and Result

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

class Student {
private:
    int rollNo;
    string name;
    float marks[3]; // Marks for 3 subjects
    float totalMarks;
    float percentage;
    string result;

public:
    void acceptDetails() {
        cout << "Enter Roll Number: ";
        cin >> rollNo;
        cout << "Enter Student Name: ";
        cin.ignore();
        getline(cin, name);
        totalMarks = 0;
        cout << "Enter marks for 3 subjects (out of 100):\n";
        for (int i = 0; i < 3; i++) {
            cout << " Subject " << (i + 1) << ": ";
            cin >> marks[i];
            totalMarks += marks[i];
        }
        calculateResult();
    }

    void calculateResult() {
        percentage = totalMarks / 3.0f;
        if (percentage >= 40.0f) {
            result = "PASS";
        } else {
            result = "FAIL";
        }
    }

    void displayDetails() const {
        cout << "\n================ STUDENT REPORT ================\n";
        cout << "Roll Number : " << rollNo << "\n";
        cout << "Name : " << name << "\n";
        cout << "Marks : " << marks[0] << ", " << marks[1] << ", " << marks[2] << "\n";
        cout << "Total Marks : " << totalMarks << " / 300\n";
        cout << "Percentage : " << fixed << setprecision(2) << percentage << "%\n";
        cout << "Result : " << result << "\n";
        cout << "================================================\n";
    }
};

int main() {
    Student s;
    s.acceptDetails();
    s.displayDetails();
    return 0;
}
```